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Fusing the RGBD SLAM with Wheel Odometry

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Abstract: This paper deals with data fusion of existing visual SLAM algorithm and wheel odometry. The result of this connection is the possibility of suppressing measurement error of each position estimation method and creating more accurate 3D model of the examined environment. We have made a brief review of existing visual SLAM projects that are available in open-source domain and as the best option we have choose the Elastic Fusion project and we enriched the existing localization pipeline with another data source that is realized by our high-precision odometry vehicle frame. The modified technique has been validated on three different scenarios. The result is the more robust SLAM algorithm and better precision 3D model of sensed environment compared to original one.

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Keywords: RGBD camera, SLAM, Kinect, Odometry, Data fusion, Dead reckoning, 3D map, Dynamic mapping

1. INTRODUCTION

These days, there are many ways how to navigate robots in outdoor environment, for example satellite systems such as GPS, Galileo, Glonas or their assisted versions with utilization of cellular phones network or Wi-Fi based localization. All these systems can determinate robots position with accuracy up to a few meters. By using the high accuracy of GPS signals with and special functions like Real-Time Kinematics (RTK), we are able to estimate global position with precision of several centimetres.

For indoor navigation, with no access to satellite signal, there have been already developed many Visual Simultaneous Localization and Mapping (SLAM) techniques. These implementations aim on creation of a type of artificial intelligence, which would be able to see and understand its surroundings with usage of inexpensive cameras, like the cheap RGBD Kinect sensor, etc. An output of these algorithms are real-time constructed 3D models of robots surroundings and its estimated position inside of these models.

Current state-of-the-art technology, like Schneider et al. (2018) or Engel et al. (2015), still has a strongly limited range of application. These days implementations are able to map areas on scale up to several hundred of square meters. The most dramatic problem is that with increasing size of the mapped area there is directly proportional amount of data that the computer has to manage.

This paper aims on extending the existing solution of pure visual odometry by using the data from more accurate sensors mounted on the mobile robot and trying to create mobile unit, which would be able to explore unknown environment with higher precision than then original solution.

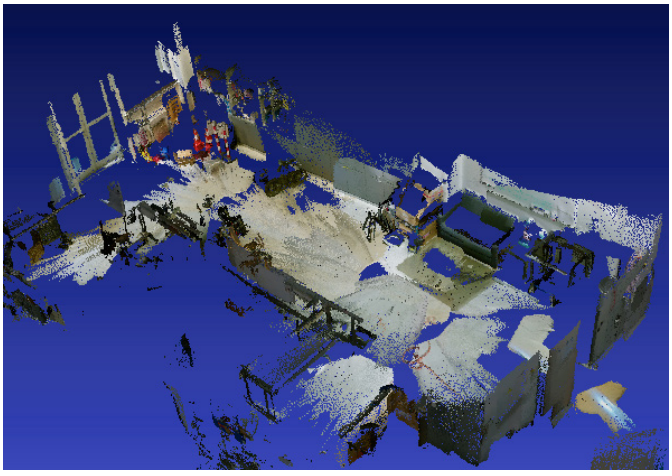


Fig. 1. An example of RGB colorized point cloud created by our modification of the Elastic Fusion software by Whelan et al. (2015b).

2. RELATED WORKS

The entire work on this experiment has been strongly motivated by the focus of our Robotics and AI research group, that is developing the robotic systems with high level of autonomy and human independent behavior. The great example could be the ATEROS robotic system created by Zalud (2006), which is primarily designed to operate in environments which are dangerous for human beings, such as radiation or chemical contaminated areas called chemical, biological, radiological and nuclear (CBRN) defense. ATEROS consists of more types of robots to cover different types of mission requirements. Currently, this system is successfully deployed in military sector.

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To make the system even more autonomous there is need to provide the ATEROS robots (like Orpheus described in Zalud (2004)) with the information about their surrounding environment that would be used in more complex mission planning algorithms or data acquisition. This should improve environment visualization for operator who is supervising the entire mission.

For the purpose of our work we have studied several RGBD SLAM algorithms, like Kinetic Fusion by Newcombe et al. (2011), Kintinuous by Whelan et al. (2015a) and Elastic Fusion described in Whelan et al. (2016), for which we have compared their computational and sensory hardware requirements, precision of the output models, environment scaling factor and the source code availability, so we would be able to modify current solution.

Finally the Elastic Fusion project has been chosen as the best candidate because of fully open-source code `https://github.com/mp3guy/ElasticFusion` and perfect documentation compared with other projects that has been non-documented or closed source at all.

3. ELASTIC FUSION

This project described in Whelan et al. (2015b) and Whelan et al. (2016) has been created by the same authors as Kintinuous, but the way, how the SLAM problem is being solved is different in many ways. The most interesting change is, that Elastic Fusion leaves the concept of Graph Based SLAM Grisetti et al. (2010), which has been the state-of-art approach for nearly two decades.

This project has been chosen as base of our work, because it introduces a new and interesting ways of solving SLAM problems and it is in active development process, so the future improvements can be expected. Another purpose is, that this program gives very precise output models in an example demo application.



Fig. 2. Example of "Dense model" created by Elastic Fusion. It is composed of 4,5mil surfels. Image comes from Whelan et al. (2015b)

The Elastic Fusions model, example shown on Fig 2, is defined as a unordered set of surfels, where each surfel is defined by its position $p \in \mathbb{R}^3$, normal $n \in \mathbb{R}^3$, color $c \in \mathbb{R}^3$, weight $w \in \mathbb{R}$ and radius $r \in \mathbb{R}$. Additionally, every surfel has its internal state (active/inactive). The surfel is active in the moment of creation and keeps active until the RGBD sensor is able to see it. If surfel has not been seen for defined period of time it comes inactive.

Interesting improvement is in the fact that local position estimation is performed only against active surfels. These are the ones that had been tracked during short period of time. In this way, the local operation window is created, so the frame-to-model process of local position estimation do not have to handle large amount of data.

There are separated processes to estimate position from RGB and depth domains and after that both pipelines are linearly combined together into single output model. In depth domain, the well-known ICP algorithm is used and in the color domain authors are minimizing the square difference of two incoming RGB frames. The system tries to figure out the small homogeneous transformation between both positions in which the images have been taken and later it minimise the sum of square differences of the original image and the second image, that is reprojected into the position, where previous image has been taken.

The main innovation comes in the field of global graph deformation. There is no background graph that would be processed during the entire run-time. Instead, for each new frame the entire new graph is constructed. It is created by randomly picking out several surfels from global model and each of those key-surfels is connected to its four neighbors. As the key-surfels are picked randomly from whole set of existing ones, the samples are taken uniformly through entire timeline of surfels creation timestamps, so the whole model is covered. This graph is used in next step to perform local and global loop closing.

During the loop closing at first the global loop closing is performed. Global loop closing is performed as follows. There are randomly selected views over the entire model and the current view is compared to selected ones. To improve performance, the views are down sampled to 80x60. If a loop is detected, the deformation is done and it distributes through the whole non-rigid model.

If no global loop closing has been detected, the local loop closing is performed. It is done only above the set of currently active surfels and it helps to keep currently visible part of model consistent. Local loop closing works in the way that transformation between active surfels and visible surfels is done. If the transformation exceeds the defined threshold, the model is deformed.

4. WHEEL ODOMETRY

Dead reckoning is a method that measures the traveled distance by continuously integrating vehicle's speed or wheels angular position change.

$$\boldsymbol{x}_k = \sum_{i=1}^{k} \Delta \boldsymbol{x}_i = \sum_{i=1}^{k} \Delta \dot{\boldsymbol{x}}_i * \Delta t \tag{1}$$

Let us imagine a robot, that lives in a one dimensional space and it moves in one direction with specified speed. If we assume, that at the beginning of a measurement the robot was in the space coordinate 0, than the position in time t is expressed as the integration of the velocity in time, or the movement can be calculated directly from wheels position change. The method is very precise, but during the time there is negative effect of accumulating a

small error, which causes large drift during long period of time. The main sources of error are wheel rotation measurement error and wheel skid.

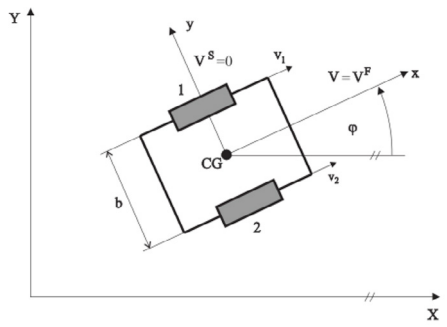


Fig. 3. Differential chassis model as described in Solc and Sembera (2008).

For the purpose of testing our results we have created the experimental high precision odometry vehicle Fig. 4. It has two wheels on the main axis, both with RI58-O/5000AS.41RB quadrature encoders mounted on every wheel. Furthermore, the vehicle is equipped with STM32F4 Discovery module, which is fitted into the extending shield. This computing unit works as a collector of the data from the encoders, which are then passed to the PC through the IP network.

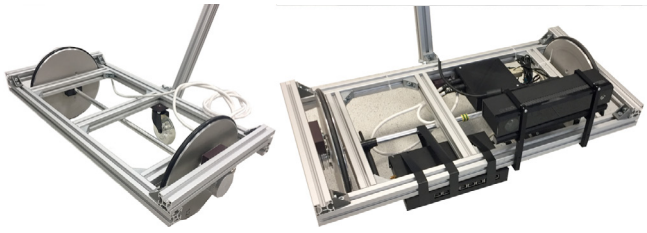


Fig. 4. The high-precision odometry vehicle used for our experiment. The trajectory estimation error of the model on the smooth surface is below 0.1% of the traversed distance. On the left the bare frame, on the right the frame equipped with Kinect v2 sensor, STM32F4 development kit in the black box behind the Kinect and the IP switch for communication between microcontroller and the PC.

The PC is running a background process, which performs a model calculation of the vehicle odometry according to the expressions 2, 3 and through a buffers queue it provides output of dead reckoning odometry to the visual SLAM thread, which is able to reduce cumulative position estimation error.

$$\boldsymbol{d'} = f(x, y, \Phi, \Delta s_r, \Delta s_l), \tag{2}$$

$$\boldsymbol{d'} = \begin{bmatrix} x \\ y \\ \Phi \end{bmatrix} + \begin{bmatrix} \frac{\Delta s_r + \Delta s_l}{2} cos(\Phi + \frac{\Delta s_r - \Delta s_l}{2b}) \\ \frac{\Delta s_r + \Delta s_l}{2} sin(\Phi + \frac{\Delta s_r - \Delta s_l}{2b}) \\ \frac{\Delta s_r - \Delta s_l}{b} \end{bmatrix}, \tag{3}$$

where $\boldsymbol{d'}$ is derivation of the robot's position coordinates x and y and orientation Φ, Δs_r and Δs_l are the distance traveled by the right and left wheels.

5. DATA FUSION AND RESULTS

5.1 Data Fusion Pipeline

The main idea of this paper is to fuse the differential chassis odometry with visual SLAM process to improve accuracy of this method. We have ported the Elastic Fusion from original Asus RGBD camera to faster and more precise Kinect v2 sensor. Also the we have performed the entire calibration process for this device, as it has been described in Lachat et al. (2015).

To fuse the vehicle odometry data into the Elastic Fusions visual position estimation process, we have extended the existing code by simple interface for external data input/output. For every single camera frame the actual 6DoF position can be read out and also for each frame the same 6DoF position can be setted into the SLAM algorithm.

The vehicle odometry feeding pipeline works in the way shown on Fig.5. The Kinect v2 in accordance with Smisek et al. (2013) provides RGBD images with 60 fps frame rate. The first frame is left without any modifications, to let Elastic Fusion performs initial position estimation. Before the second frame will be processed, the visual odometry position is taken as well as the estimation of odometry vehicle position does. Both data packages are passed into Kalman filter that has been designed using the Faragher et al. (2012) work, where the fusion is performed and after that the newly estimated position is forced into Elastic Fusions and vehicles odometry pipelines. This process is looping during entire SLAM run-time.

The fusion is realized by a Kalman filter, which has been proposed by Kalman (1960). We have implemented the 1D Kalman filter that is designed to be able to model filtered value (x, y and Φ), and the derivation of this value (linear and angular speed) as a inner states. Every single instance of the Kalman filter is processing the one of the inner states mentioned above (position coordinates x and y and the orientation Φ). For every new RGBD frame the SLAM algorithm estimates the position and this newly calculated 6DoF position and orientation is used as an input into the all three filters which generates full iteration of Kalman filter. In the same time the second input into the Kalman filters are the position and orientation of the vehicle estimated by wheel odometry. The filters predict the future states based on the Δt and combine the predictions with the input data. The result of the full iteration are the filtered values of x, y and Φ, which are setted into the visual odometry and wheel odometry as a finally estimated new 6DoF position.

5.2 Results

The visual odometry can handle very well the position estimation in highly rugged environment. The accuracy of wheel odometry is strongly dependent on precision of the calibration. During our experiments, have calibrated the wheel odometry with precision, of 0.2% relative error in

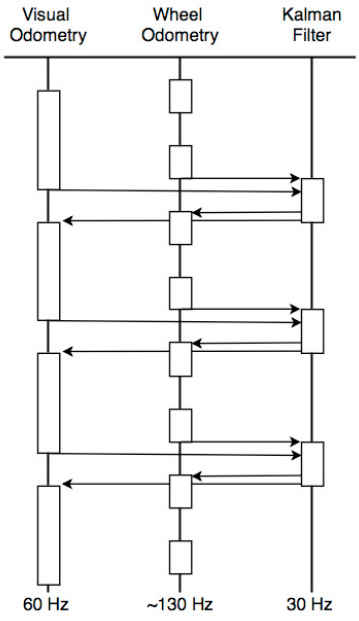


Fig. 5. Visual and odometry fusion pipeline. After every incoming frame the data from visual SLAM and from wheel odometry are merged in the Kalman filter and the results are pushed back to odometry pipelines.

linear direction movement and 1% relative error in rotation movement.

The validation experiments have been performed in three different environments. Each of them has a different characteristics, so both odometry methods have various accuracy during the 3D model construction.

The first experimental environment was the a laboratory room, see the Fig. 1. As the laboratory contains large number of various furniture, devices and other stuff, it creates a highly contrast environment in which even visual odometry gives very good results. Also loop closing, by example in Bazeille and Filliat (2010), works very well. In this case wheel odometry does not improve SLAM results in any useful way.

The second experiment has been conceived as a pass between two laboratories through short section of corridor (see the Fig. 6, when direction was changed two times. The hardest part for SLAM was to orientate itself during the corridor pass through, because there are no contrast shapes. The pure visual SLAM usually gets lost when the robot rotates in a low contrast environment. With fusion of both data sources we have been able to pass the corridor without losing orientation.

The main benefit in this experiment shows, that operator, that controls robot with installed SLAM does not need to take care about ruggedness of environment, and if visual odometry gets lost, the position estimation by wheel odometry still works, until visual odometry fixes the positioning. The model of the second experiment can be seen on the Fig. 6.

The third experiment was a straight pass through approximately forty meters long corridor, see the model on the

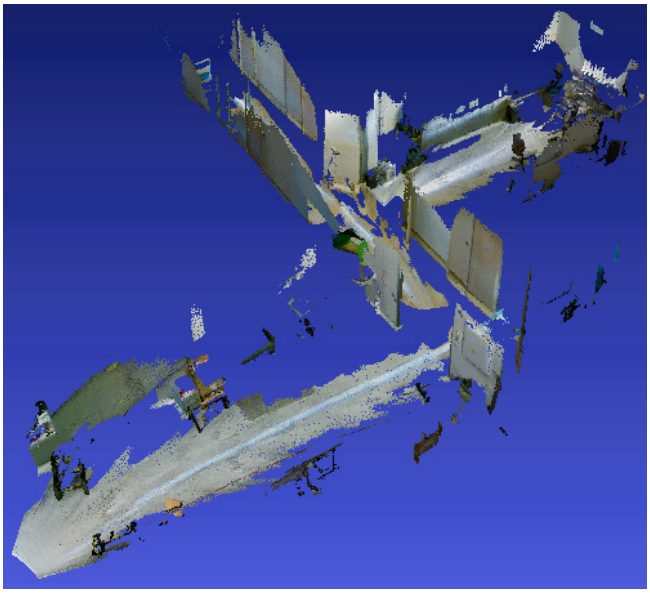


Fig. 6. Second experiment visualization. The session began in the laboratory on the bottom at the figure. The measurement vehicle traveled through the doors at the corridor, there it turns left by 90 degrees, move forward and after several meters it turns right and entered the second laboratory. The biggest challenge for the raw visual SLAM was the first turn where it usually failed. The combined method outperform the original solution.

Fig. 7. Also, in this experiment the most interesting results were achieved.

The wheel odometry is very accurate during small scale measurement, but has disadvantage of continuous integrated drift from ground truth. On the other hand, the visual odometry has large uncertainty of current position estimation, but because of close-loop feedback it is able to compensate the mentioned wheel odometry drift.

In detail, the experiment of corridor length measurement has been performed in following way. SLAM was started by looking on the first doors and slightly moved backward, to established 3D model of near surrounding. After this, it returned to the original position and turned around by 180°. Usually in this moment the pure visual odometry gets lost as it was focusing on the non-contrast white wall and the position estimation from RGB and depth cameras failed. In the same conditions, with both methods combined together the position estimation worked highly robust and was able to keep tracking the surroundings. In the last part of the experiment the vehicle with SLAM was moving straight forward to the end of the corridor with velocity of about 0.5 ms^{-1}. At the end of the corridor the measurement has been terminated and results has been saved.

The following image Fig. 7 shows the difference between visual odometry based 3D model construction and 3D model based on fused data. It is easy to see, that the pure visual odometry model exhibits large amount of geometrical distortion, while model constructed by fused data SLAM has much better geometrical expression of entire straight corridor.

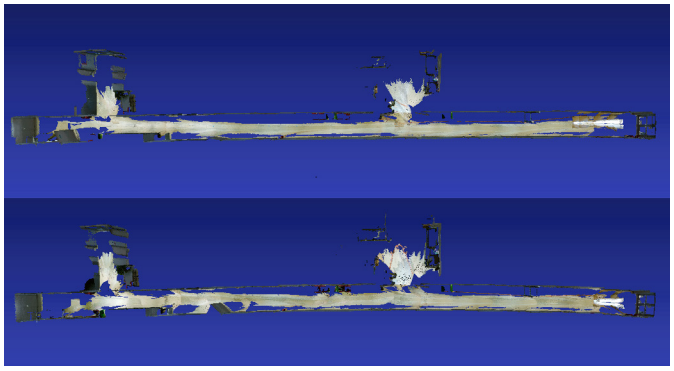


Fig. 7. Bird view of the 3D models distortion comparison. Top - fused method, bottom - raw visual odometry. The model build by raw visual odometry shows significantly bigger geometrical distortion compared to the combined method.

There are several ways of SLAM accuracy estimation as described in Endres et al. (2012). One of possible criterion to determinate accuracy of the SLAM method could be expression of relative error of straight corridor measurement.

The ground truth value of the corridor length was measured from the beginning of the corridor to the very opposite side of corridor parallel to the ground. This ground truth value has been established by the Lecia DISTO D8 laser scanner, that has measurement accuracy 1 mm.

The relative error of the experiments has been expressed as follows:

$$\delta = \frac{M - S}{S} * 100[\%], \tag{4}$$

where M is measured value and S is ground truth value.

Method	Measured Length [m]	Relative Error [%]
Visual odometry	42.30	2.70
Wheel odometry	43.02	1.04
Fused data	43.29	0.41
Ground truth	43.47	-

Table 1. Comparison of the relative error of the corridor model dimensions constructed by the different odometry methods.

From the Table 1 we can assume, that the results of Elastic Fusion with additional data source achieve several times better results (6 times lower relative error) than the original solution does. The wheel odometry helps to keep high accuracy on short time scale and visual SLAM kept the wheel odometry drift at a low value, even on a quite long distance.

6. CONCLUSION

This paper deals with visual SLAM and the method to improve its accuracy. The entire work is based on the open-source project Elastic Fusion, which was modified and the results were verified in several validation experiments.

We have forked the mentioned open-source SLAM project called Elastic Fusion and run it on Ubuntu 14.04 with



Fig. 8. 3D models geometrical distortion, top - fused method, bottom - raw visual odometry. The raw method shows significant distortion in the right-hand direction along to the forward axis.

Kinect v2 RGBD camera sensor and we have implemented the data fusion between visual odometry and the wheel odometry, based on Kalman filtration.

The final results show the achieved improvements. The accuracy improvement has been tested in three different scenarios. The most interesting result was the one, when SLAM had to scan forty meters long corridor. The pure visual SLAM shows 2.7% relative error the reconstructed corridor model dimensions. The pure wheel odometry show 1% relative error and fused method creates 3D model with 0.4% relative error. It means over six times smaller measurement error compared to the original pure visual SLAM.

The next possible extension could be done in field of virtual reality. The entire 3D constructed model could be rendered into two separated displays, each one for a single eye, so the illusion of 3D space could be created. With virtual reality headset the operator, that controls the robot on its mission could get much better experience during exploration of unknown areas and would be able to comprehend much better, how the robot is situated in its environment.

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